

3E1206	Roll No. _____	[Total No. of Pages : 3]
	3E1206	
	B.Tech. III-Sem. (Main/Back) Examination, January - 2025	
	Automobile Engg.	
	3AE2-01 Advance Engineering Mathematics-I	
	AN, AG, AE, CE, CR, EC, EI, ME, MH, PT	

Time : 3 Hours

Maximum Marks : 70

Instructions to Candidates:

Attempt all Ten questions from Part A, Five questions out of Seven questions from Part B and Three questions out of Five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

PART - A

(Answer should be given up to 25 words only)

All questions are compulsory.

(10×2=20)

1. What is the value of $\Delta^n(e^x)$; if $h = 1$.
2. Write the trapezoidal rule.
3. Write the formula for Euler's modified method.
4. Find the first approximation to a real root of equation $x^3 - 3x^2 - 2 = 0$ by Regula falsi method between 3 to 4.
5. What are the existence conditions of Laplace transform?
6. Find $L^{-1}\left[\frac{1}{s^2}\right]$
7. Define Fourier sine and cosine transforms.

8. Find the Fourier transform of $f(t) = 1, |t| < 1$.
9. Find the Z - transform of the sequence $u_n = \{2, -4, 6, -8\}$.
10. Find $Z^{-1}\left[\frac{z}{z-3}\right]$.

PART - B

(Analytical/Problem solving questions)

Attempt any Five questions.

(5×4=20)

1. Using Lagrange's interpolation formula, find the value of $y = f(x)$ at $x = 9.5$ for the following:

x	7	8	9	10
$f(x)$	3	1	1	9

2. Compute the value of $\int_0^6 \frac{dx}{1+x^2}$ by Simpson's " $\frac{1}{3}$ " rule.
3. Using Euler's method, solve the differential equation $\frac{dy}{dx} = x + y$; with initial condition $y(0) = 1$ for $x=1$, using $h = 0.2$
4. Find Laplace transform of $\sin \sqrt{t}$
5. Find $L^{-1}\left[\frac{4s+5}{(s+2)(s-1)^2}\right]$
6. Find the Fourier Sine and cosine transform of $f(x)$, where $f(x) = \begin{cases} 1, & 0 < x < a \\ 0, & x > a \end{cases}$
7. Using convolution theorem, evaluate $z^{-1}\left[\frac{z^2}{(z-1)(z-3)}\right]$

PART - C

(Descriptive/Analytical/Problem Solving/Design question)

Attempt any Three questions.

(3×10=30)

1. Use Stirling's formula to find y at $x = 12.2$ from the following data:

x :	10	11	12	13	14
y :	23967	28060	31788	35209	38368

2. Calculate $y(0.2)$ using Runge - Kutta fourth order method to solve $\frac{dy}{dx} = x + y^2$, $y(0) = 1$, $h = 0.2$

3. Use Laplace transform to solve $(D^2+25)y = 10 \cos 5x$, given that $y(0) = 2$, $y'(0) = 4$.

4. Find the complex fourier transform of $e^{-|x|}$

5. Use Z transform to solve the difference equation. $U_{n+2} + 4U_{n+1} + 3U_n = 3^n$, given that $u_0 = 0, u_1 = 1$.
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